

Minh Hai NGUYEN

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I am a second-year PhD candidate in MAMBO team at Toulouse university (expected graduation: Dec. 2026), advised by Prof. Pierre Weiss and Prof. Edouard Pauwels. My research explores generative models for inverse problems in imaging, with applications in microscopy.

EDUCATION

PhD in Applied Mathematics

Toulouse University, France

Advised by Prof. Pierre Weiss and Prof. Edouard Pauwels

Research topics: Inverse Problems in Imaging, applications to Microscopy

2023 -

Master of Engineering in Applied Mathematics

INSA Toulouse, France

Major in AI / Data Science

(*Location: 2018-2020 in Vietnam and 2020-2023 in France)

2018 - 2023

EXPERIENCE

Research Engineer

Agenium Space

Correcting micro-vibrations in satellite imaging with pushbroom cameras.

Dec. 2022 - Sept. 2023

Research Intern

Institut Mathématiques de Toulouse

Product-convolution neural networks: applications to image deblurring with space-varying blurs.

Jun. 2022 - Sept. 2022

PUBLICATIONS

- [1] M. H. Nguyen, E. Pauwels, and P. Weiss. "How Diffusion Prior Landscapes Shape the Posterior in Blind Deconvolution". In: *Pre-print* (2024).
- [2] M. H. Nguyen, F. de Vieilleville, and P. Weiss. "DeepVibes: Correcting Microvibrations in Satellite Imaging With Pushbroom Cameras". In: *IEEE Transactions on Geoscience and Remote Sensing* 62 (2024), pp. 1–9.
- [3] M. H. Nguyen and P. Weiss. "Comparing Plug-and-Play and Unrolled networks". In: *Pre-print* (2024).

LANGUAGE

English: fluent

French: fluent

Vietnamese: native

PROGRAMMING LANGUAGE

Python (advanced): PyTorch (DDP, AMP), Jax, scientific computing

C/C++ (intermediate)

OTHER SERVICES

Open source project

DeepInverse

PyTorch library for solving imaging inverse problems using deep learning

Github Repository

Teaching

- Signal Processing (1st Master level) at INSA Toulouse, in 2024.

- Stochastic Optimization and Learning (1st Master level) at INSA Toulouse / ENSEEIHT, in 2024.

HONOR AND AWARDS

French National Prize for Open Science Library for DeeplInverse.

Documentation prize

By Ministère de l'Enseignement Supérieur et de la Recherche

Nov. 2024

Doctoral Research Fellowship, French Ministry of Higher Education and Research

Fully Funded Doctoral Research Grant (2023-2026)

2023

IA Pau Competition

Best project award

By IA Pau organization, funded by Boavizta (1500eur prize).

2023